

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2084

Mickens 138 kV
March 16, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2084. The Interconnection Customer proposes a 89.4 MW solar photovoltaic facility interconnecting at the Mickens 138 kV substation in the Virginia area. The request was received on May 15, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 738 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$165,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Thole St / Diamond Springs 138 kV: 109.0% loading (211 MVA) following the outage of the Newprt News / Mickens 345 kV; DFAX = 6.1%.

For the loss of Wheelabrator Portsmouth / Victory Way 138 kV, the Ruritan / Gosport 138 kV loads to 101.6% of its 230 MVA rating. Project DFAX: 5.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Thole St / Diamond Springs 138 kV	Reconductor line	\$1,565,000	4% (\$64,000)
Gosport Substation / Rritan 138 kV	Reconductor line	\$1,331,000	8% (\$101,000)
	Total allocated to project		\$165,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.